

# A Bayesian Analysis of the Marth-Fox Matchup in Super Smash Bros. Melee

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# **Chapter 1**

## **Introduction and Motivation**

## 1.1 Introduction

In Super Smash Bros. Melee (colloquially known as just “Melee”) for the Nintendo Game-Cube, there are 26 characters to choose from, including fan-favorites like Mario and Luigi, to less known characters like Ness and Ice Climbers. In competitive Melee, each player has four lives, also known as stocks. A player wins by depleting all of their opponents stocks before losing theirs.

Typically, in competitive games or sports, each individual or team has equal access to the same moves and techniques. For example, in chess, each player has the same number and types of pieces. In game theory, this is an example of a symmetric game, in which the players have the same strategy set [1] p. 186. However, in fighting games, each character has a unique moveset with individual strengths and weaknesses. This is an example of an asymmetric game, in which each character has a unique strategy and set of options that can be employed. Certain characters have swords or projectiles, allowing them to attack opponents from a distance. Some characters are fast, but easy to knock off the stage, while others are slow, but more resilient to being defeated.

Additionally, Melee is unique from other fighting games, in that it is a platform fighting game. Typically, fighting games are played on a two-dimensional plane with no frills. The goal is to deplete the health of your opponent. In Melee, the goal is to knock your opponent off of the stage to deplete their stocks. Each stage has unique properties that can affect how certain characters play. Characteristics such as platforms, stage transformations, and other interactable elements affect how players approach matchups and can directly affect gameplay. It is agreed upon in the community that certain stages are better for certain characters due to the tools available to the character.

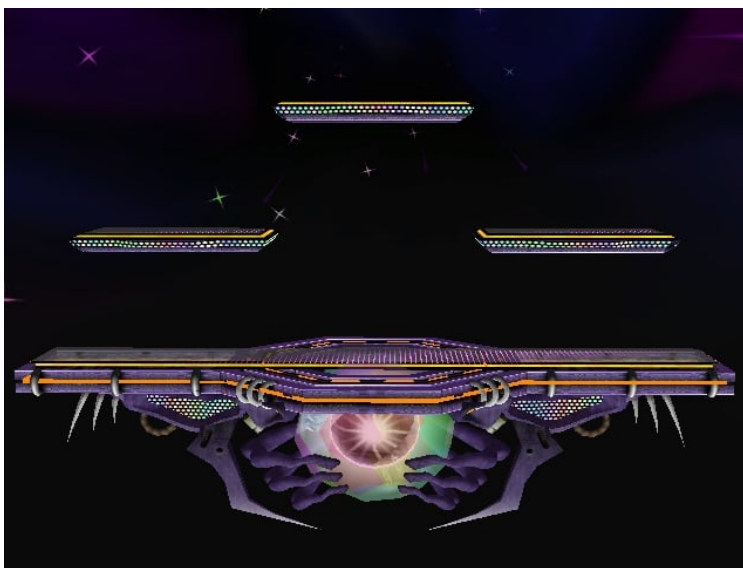


Figure 1.1: Battlefield, a stage that offers 3 platforms to extend or escape from combos.

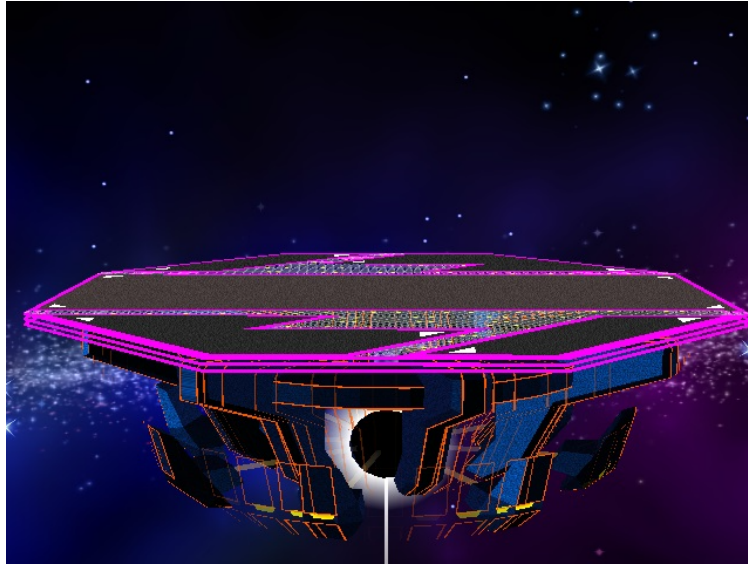


Figure 1.2: Final Destination, a stage without platforms.

Because of the asymmetric nature of Melee's characters, it is natural to assume that some characters perform better against others due to having a superior toolkit in a given matchup, or they may perform better on more stages than their opponent. To express how strong a character is, players will create a "matchup chart" or "matchup spread" for the character. This is a measurement of a character's probability of winning against other characters in the roster. When designing a matchup chart, it is typically assumed that the players of each character are of equal skill level. Matchup charts are also typically designed based on the perception of how characters perform against one another, given their respective toolkits and performance on certain stages. There is typically little to no formal statistical analysis when designing these spreads.

While some matchup charts are expressed using  $\pm 1, 2, \dots$ , this is not as descriptive as to how favorable or unfavorable a matchup may be. Explicit probabilities may also be used when describing a matchup. Formally, let us define a matchup of character A against character B. When discussing this matchup, we would say "Character A's matchup against Character B is  $x/y$ ," where  $x$  is the probability that character A would win,  $y$  is the probability that character B would win, such that  $x + y = 100$ . For example, an even matchup would be designated as "50/50". A slightly favorable matchup would be "55/45", while a heavily unfavorable matchup may be expressed as "35/65".

|                                   |                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------|
| Super Favored<br>(65+:35-)        | LUIGI, ANON, DOR, YOUNG, LINK, PICHU, NESS, KIRBY, BOWSER                                      |
| Favored<br>(6:4)                  | PEACH, ICE, CLIMBERS, YOSHI, DR. MARIO, SAMUS, D K, MARIO, LINK, MR. GAME & WATCH, MENTWO, ROY |
| Slight Favored<br>(55:45)         | FOX, FALCO                                                                                     |
| Basically Even<br>(52.5:47.5 LOL) | C. FALCON, ZELDA, JIGGLY PUFF                                                                  |
| Even                              | MARTH                                                                                          |
| ?                                 | PIKACHU                                                                                        |

Figure 1.3: A Marth matchup chart, made by top Marth player KoDoRiN

## 1.2 Motivation

A tier list is an ordered list of characters, where each character is ranked based on how strong they are relative to the rest of the cast of characters. Tier lists are often subjective, and they will change over time as the strategies for each character develops over time.

| Super Smash Bros. Melee Tier List #13                                               |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| S                                                                                   |                                                                                     |                                                                                     |                                                                                     | A                                                                                   |                                                                                     |                                                                                     | B+                                                                                  |                                                                                     |                                                                                     |                                                                                      | B-                                                                                    |                                                                                       |
| 1                                                                                   | 2                                                                                   | 3                                                                                   | 4                                                                                   | 5                                                                                   | 6                                                                                   | 7                                                                                   | 8                                                                                   | 9                                                                                   | 10                                                                                  | 11                                                                                   | 12                                                                                    | 13                                                                                    |
|  |  |  |  |  |  |  |  |  |  |  |  |  |
| 1.68                                                                                | 2.36                                                                                | 3.18                                                                                | 3.56                                                                                | 4.66                                                                                | 5.82                                                                                | 6.84                                                                                | 8.74                                                                                | 9.62                                                                                | 9.69                                                                                | 10.11                                                                                | 12.23                                                                                 | 12.61                                                                                 |
| C+                                                                                  |                                                                                     | C-                                                                                  |                                                                                     |                                                                                     |                                                                                     | D                                                                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                      | F                                                                                     |                                                                                       |
| 14                                                                                  | 15                                                                                  | 16                                                                                  | 17                                                                                  | 18                                                                                  | 19                                                                                  | 20                                                                                  | 21                                                                                  | 22                                                                                  | 23                                                                                  | 24                                                                                   | 25                                                                                    | 26                                                                                    |
|  |  |  |  |  |  |  |  |  |  |  |  |  |
| 14.83                                                                               | 15.53                                                                               | 16.42                                                                               | 17.31                                                                               | 17.66                                                                               | 17.95                                                                               | 20.22                                                                               | 21.63                                                                               | 22.07                                                                               | 22.78                                                                               | 23.49                                                                                | 24.26                                                                                 | 25.74                                                                                 |

Figure 1.4: The 13th community Melee tier list. Fox and Marth occupy the first and second spots, respectively.

One of the most prominent matchups in competitive melee is Marth vs. Fox. Both of these characters have occupied the top spots of almost every competitive tier list for over two decades. Both characters have been extremely successful, and many players have piloted the characters to major tournament victories. They each have strong toolkits, and the ability to perform well against any character in the game. However, there has been debate in the Melee community about how Fox and Marth perform against each other.

Most players would argue that the matchup is even. Each character has tools in their moveset to exploit the other character's weaknesses. However, top Melee players have expressed different

opinions. The most infamous claim was made by the player William “Leffen” Hjelte. Hjelte, a Fox player, argued that Marth is favored in the matchup 60/40, and it became somewhat of a joke in the Melee community. When referencing the matchup for Marth against Fox, he claimed “6-4 is probably [right], both according to results and ... Marth [having] a way easier time improving in that matchup” [2]. Although Hjelte is arguably one of the best Fox players of all time and has incredible knowledge about Melee, many claimed that such a matchup spread was incredibly far-fetched.

I thought that this debate would be interesting to approach from a Bayesian lens. Given expert prior beliefs on the matchup, we could use results from individual matches to determine a more statistically sound matchup spread for a character. Furthermore, Melee has many characters that get less usage due to being considered weaker than most choices. Thus, there is little data for some of these more obscure matchups. Bayesian methods thrive in this environment, allowing us to define such matchup spreads for less-used characters.

# **Chapter 2**

## **Data Collection**

## 2.1 Slippi

Melee was originally released in 2001 with limited features. [Slippi](#), released in June 2020, is a project designed to bring features of modern-day e-sports to Melee. These include portable replay files, complex gameplay stats, online matchmaking with improved online netcode, and more [3]. Not only did this make Melee more accessible through the introduction of online play, a feature missing from the original game, but Slippi also ushered in an era of data analytics surrounding Melee. Using Slippi game data, it is possible to easily analyze match data without having to manually record match results.

The data used in this project was sourced from [Chart.slp](#), a platform that hosted a large corpus of Slippi match data. Although it is no longer maintained, it is possible to download a backup of the database, containing data for 5,594,571 different matches. This provides sufficient data for Bayesian analysis across nearly every matchup in Melee.



## **Chapter 3**

# **Modeling the Marth-Fox Matchup**

### 3.1 Model 1: A Simple Binomial-Beta Conjugate

For the first model, we will approach it completely analytically. We will model the win rate of a character in a matchup, without factoring in stage choice. Let  $\theta$  represent the probability of a character winning a given matchup. Let the likelihood of our model be represented as a binomial distribution  $x_i|\theta \sim \text{Bin}(n, \theta)$ , where  $n$  represents the number of matches in the sample. A win for the character would be recorded as a 1 (success), while the loss would be recorded as a 0 (failure). We can specify a Beta prior generally as  $\theta \sim \text{Be}(\alpha, \beta)$ . A Beta prior is flexible, as it allows us to experiment with different initial beliefs for a given matchup.

The Beta prior can form a conjugate distribution with the posterior with respect to the Binomial likelihood. This enables us to find a fully closed-form posterior. Using existing conjugacy tables [4] p. 341 gives the posterior  $\theta|X \sim \text{Be}(\alpha+x, \beta+n-x)$ . The mean of the posterior,  $\frac{\alpha+x}{(\alpha+x)+(\beta+n-x)}$ , gives us the expected win rate for a character in a given matchup.

For the Marth-Fox matchup, assume that we are modeling Marth's win rate against Fox. Below are graphs representing the posterior matchup distribution, given the extracted Slippi match data and varying priors.

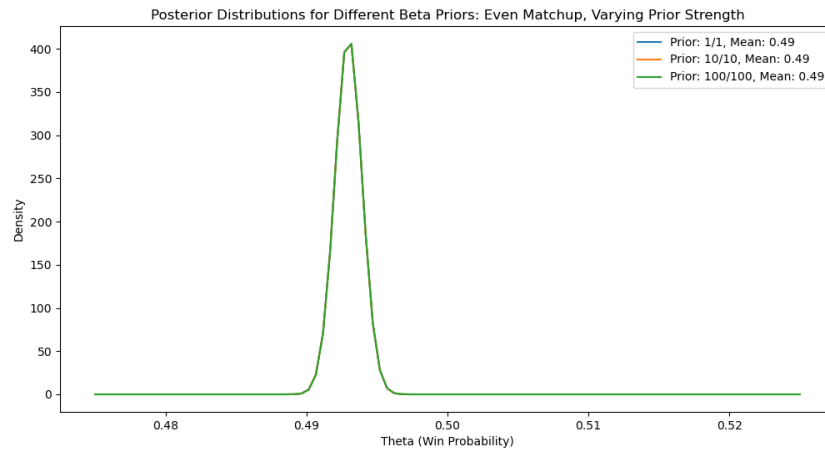


Figure 3.1: Modeling the Marth-Fox matchup using an even (50/50) prior with varying strengths.

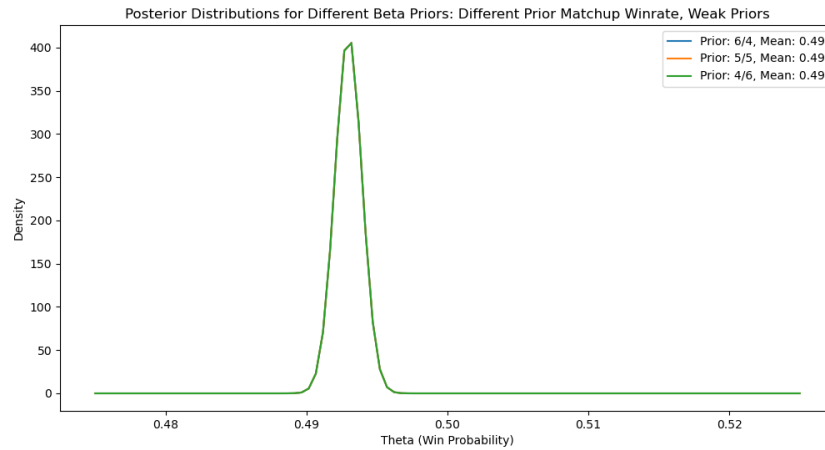


Figure 3.2: Modeling the Marth-Fox matchup with varying, weak priors.

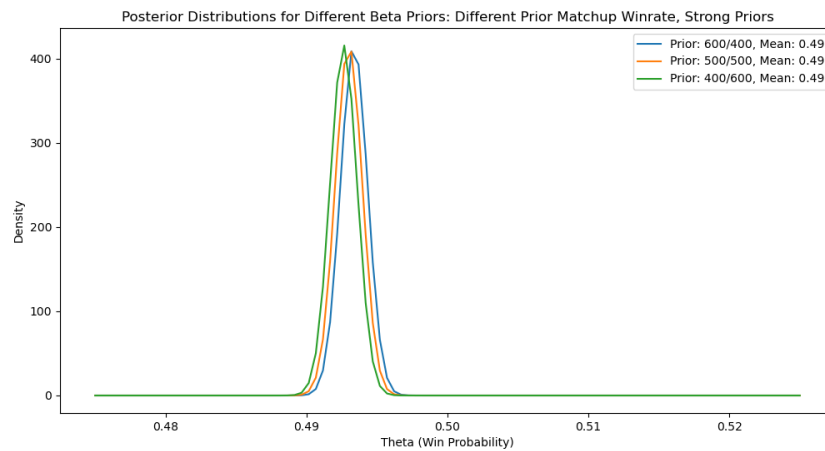


Figure 3.3: Modeling the Marth-Fox matchup with varying, strong priors.

As shown in these graphs, the expected win rate of Marth against Fox is greatly impacted by the observed data ( $\approx 270,000$  matches observed). This makes sense because the posterior distribution parameters will be greatly influenced by the likelihood. Thus, in the case of a large data sample like this, the impact of the priors, regardless of perceived strength, is rendered nearly meaningless due to the sheer amount of observed data. However, for cases of uncommon matchups, where there may not be a lot of available data, the Bayesian model will allow our prior distribution to be incorporated into the posterior to a greater degree. Below are graphs of the posterior when, in a hypothetical sample of 10 Marth-Fox matches, Marth wins 8 of them.

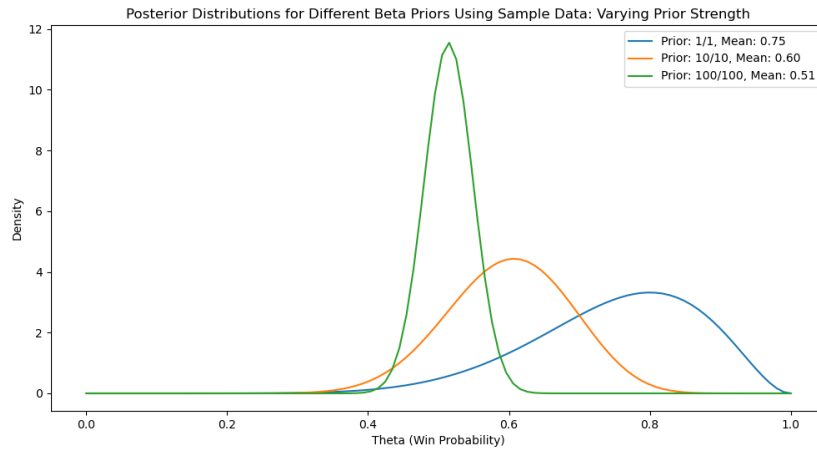


Figure 3.4: Modeling the Marth-Fox matchup using an even (50/50) prior with varying strengths using sample data.

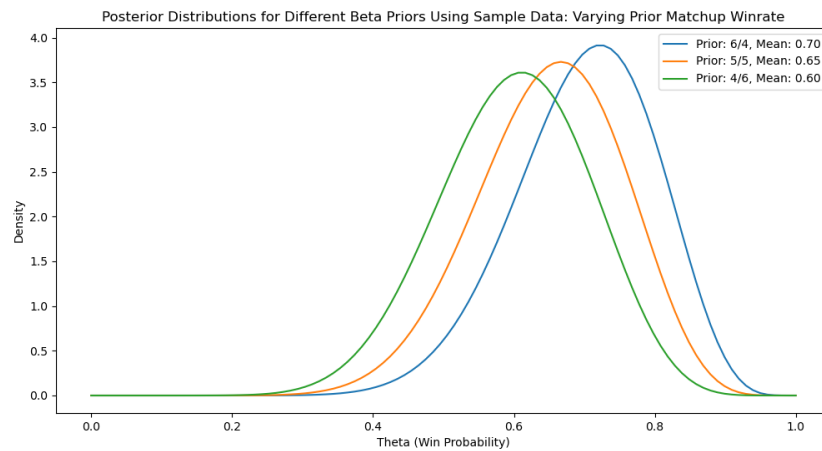


Figure 3.5: Modeling a different Marth-Fox expected win rates given varying matchup priors of similar strength using sample data.

Looking at the above figures, it is more evident how changing a prior's strength or shape will affect the posterior distribution. Again, in the case of Melee, given  $n = 26$  characters, there are  $\frac{n(n+1)}{2} = \frac{26 \cdot 27}{2} = 351$  total unique matchups. By leveraging a Bayesian approach, we can more accurately model matchup probabilities for characters with significantly lower usage in competitive settings.

Although this is a good exercise and a simple way to use Bayesian methods to better model matchups in Melee, there are several components not included in this initial model. First, as previously explained, stages in Melee often make a difference in a matchup. In the Marth-Fox matchup, it is argued that Final Destination is a strong choice for Marth due to him gaining access to techniques not available on stages with platforms. Thus, we are interested in also understanding character matchups by stage. Second, it is likely that not all of the observed matches were against

evenly matched players. We may want to design a model that factors in player skill when trying to determine the true matchup spread.

## 3.2 Model 2: Marth-Fox Matchup by Stage

In competitive Melee, there are 6 tournament-legal stages: Battlefield, Final Destination, Dream Land, Fountain of Dreams, Pokemon Stadium, and Yoshi’s Story. We are interested in understanding how the Marth-Fox matchup varies by stage. In competitive Melee, before a set begins, players will go through a process known as “stage striking”, in which players take turns removing a stage from being selected until there is only one remaining [5]. It is important for a player to understand how their character performs on each stage in a given matchup when deciding which stages to strike.

To determine if there is a significant effect a stage has on the Marth-Fox matchup, we can utilize a logistic regression model. This fits since the response data (a match win or loss) is binary in nature. We define our model as

$$\begin{aligned} y_i | p_i &\sim \text{Bernoulli}(p_i), i = 1, \dots, 6 \\ \text{logit}(p_i) &= \alpha + \gamma_i y_i \\ \alpha &\sim N(0, 10) \\ \gamma_i &\sim N(0, 10) \\ \sum \gamma_i &= 0 \end{aligned}$$

where  $\alpha$  represents the base “matchup effect” and  $\gamma_i$  represents the “stage effect” on the matchup probability. A positive value for these values indicates a favorable matchup for Marth, either generally, or given a particular stage. As a note, to speed up the model, only a subset of the full corpus of data was used (10,000 matches), which contributes to the difference in values than may have been calculated in the previous section. Below are the summary statistics for this model using a 95% HDI credible interval:

|                           | mean   | sd    | hdi_2.5% | hdi_97.5% | mcse_mean | mcse_sd | ess_bulk | ess_tail | r_hat |
|---------------------------|--------|-------|----------|-----------|-----------|---------|----------|----------|-------|
| alpha                     | -0.284 | 0.020 | -0.321   | -0.243    | 0.000     | 0.000   | 5294.000 | 3306.000 | 1.000 |
| gamma[YOSHIS_STORY]       | 0.121  | 0.045 | 0.036    | 0.210     | 0.001     | 0.001   | 5943.000 | 2928.000 | 1.000 |
| gamma[FOUNTAIN_OF_DREAMS] | -0.019 | 0.046 | -0.107   | 0.070     | 0.001     | 0.001   | 5386.000 | 3267.000 | 1.000 |
| gamma[FINAL_DESTINATION]  | 0.069  | 0.046 | -0.019   | 0.163     | 0.001     | 0.001   | 6276.000 | 2805.000 | 1.000 |
| gamma[POKEMON_STADIUM]    | -0.081 | 0.043 | -0.162   | 0.004     | 0.001     | 0.001   | 5336.000 | 2909.000 | 1.000 |
| gamma[BATTLEFIELD]        | -0.019 | 0.044 | -0.108   | 0.060     | 0.001     | 0.001   | 6070.000 | 3172.000 | 1.000 |
| gamma[DREAMLAND]          | -0.070 | 0.046 | -0.161   | 0.017     | 0.001     | 0.001   | 5253.000 | 3418.000 | 1.000 |

Looking at this data, we see that the credible interval for  $\alpha$  does not cross 0. This implies that, disregarding any additional stage factor, Marth is likely to have a losing matchup against Fox. Specifically, when converting the log-odds to a probability, we estimate the base probability of Marth winning against Fox to be approximately 42.9%. This contradicts both the assumption that the matchup is even, or the possibility that the matchup is in Marth’s favor.

Analyzing the  $\gamma_i$  values, we can determine what stages could be beneficial for Marth. Yoshi’s Story appears to be the only stage in which we can conclusively determine (given the 95% HDI credible interval) that it is more likely that Marth will win. While not conclusive, Final Destination

appears to be in Marth's favor. Similarly, Dream Land and Pokemon Stadium appear to lean in Fox's favor, although this is also not conclusive. Battlefield and Fountain of Dreams appear to be relatively even for both characters. Given this data, a Marth player's choice for stage bans or selections can be more strongly statistically supported.

We also can conduct a hypothesis test to see if the belief that the matchup being 60/40 in Marth's favor holds any merit. We can do this by checking if  $\text{invlogit}(\alpha) \geq 0.6$  for each sampled  $\alpha$  in the chain. The resulting data of this hypothesis test is as follows:

|    | mean  | sd    | hdi_2.5% | hdi_97.5% | mcse_mean | mcse_sd | ess_bulk | ess_tail | r_hat |
|----|-------|-------|----------|-----------|-----------|---------|----------|----------|-------|
| h1 | 0.000 | 0.000 | 0.000    | 0.000     | 0.000     | NaN     | 4000.000 | 4000.000 | NaN   |

It can be seen that in the sampling there were no samples that fit this criteria. Thus, this is a strong indication that the belief that the Marth Fox matchup is 60/40 in Marth's favor is not supported at all by this model.

*Note: When conducting the hypothesis test, arviz returned runtime warnings due to attempting to divide by zero, resulting in NaN values for mcse\_sd and r\_hat. This is likely due to the hypothesis test returning no viable samples given the constraint  $\text{invlogit}(\alpha) \geq 0.6$ . These warnings are safe to ignore in this case.*

The downside of this model is that it does not factor in player skill. The benefits of certain stages may be emphasized more when a player has the necessary skill to exploit certain techniques. Additionally, it is possible that the matchup winrate in general may shift depending on the skill level of players. The next model will propose an approach to incorporate Slippi ratings into our model.

### 3.3 Model 3: Factoring in Elo

In newer versions of Slippi, a ranked mode was added. In this mode, players are given a rating. A higher number is indicative of a more skilled player, and vice versa. We can use this information as a factor in modeling the matchups in Melee.

Slippi follows a rating system in which players are categorized into different "ranks" based on their specific rating [6]. For instance, if a player has a rating of 1450, they would have a rank of Gold 1. This distinction is important in how Elo is factored into the model.

| Rank        | ELO Required                |
|-------------|-----------------------------|
| Bronze 1    | 0                           |
| Bronze 2    | 766                         |
| Bronze 3    | 914                         |
| Silver 1    | 1055                        |
| Silver 2    | 1189                        |
| Silver 3    | 1316                        |
| Gold 1      | 1436                        |
| Gold 2      | 1549                        |
| Gold 3      | 1654                        |
| Plat 1      | 1752                        |
| Plat 2      | 1843                        |
| Plat 3      | 1928                        |
| Dia 1       | 2004                        |
| Dia 2       | 2074                        |
| Dia 3       | 2137                        |
| Master 1    | 2192                        |
| Master 2    | 2275                        |
| Master 3    | 2350                        |
| Grandmaster | Top of Region<br>AND > 2192 |

Figure 3.6: Rating cutoffs for each rank in Slippi.

The Bradley-Terry model provides an approach in which we can model the probability of a Marth player winning a match against a Fox player given the strength of the players. Let  $\beta_i, \beta_j$  represent the perceived “strength” of the Marth and Fox player respectively, in terms of rating. The log-odds of the Marth player beating the Fox player is equal to the difference of  $\beta_i$  and  $\beta_j$  [7]. Through strategic ordering of parameters, we can also introduce other effects that can factor into the outcome of a game between two players, including the previously explored “matchup” factor and “stage” factor. Thus, we can naturally extend the previously defined model for Marth’s matchup against Fox in a vacuum, to now model the probability of a specific Marth player winning against a Fox player given their Slippi ratings.

Unfortunately, the data sampled for this project did not have access to player rating data. To prove out this new model, rating data was synthesized based on the existing range of Slippi ratings. To normalize the rating data into relative skill differences, I opted to divide the difference in player rating by 300. This aligns with the approximate rating difference between major ranks (i.e. between Silver and Gold, Gold and Platinum, etc.) across Slippi’s ranked system. Thus, for the modified Bradley-Terry model, we use a scale factor of 300 and a base of  $e$ , slightly different than the traditional setup for Elo in chess, which utilizes a scale factor of 400 and base of 10 [8].

More formally, we can define our new model in the following way:

$$\begin{aligned}
 y_i | p_i &\sim \text{Bernoulli}(p_i), i = 1, \dots, 6 \\
 \text{logit}(p_i) &= \beta \left( \frac{\epsilon_m - \epsilon_f}{300} \right) + \alpha + \gamma_i y_i \\
 \alpha &\sim N(0, 10) \\
 \beta &\sim N(0, 10) \\
 \gamma_i &\sim N(0, 10) \\
 \sum \gamma_i &= 0
 \end{aligned}$$

Where  $\alpha$  and  $\gamma_i$  are the same as previously defined,  $\epsilon_m, \epsilon_f$  is the Slippi rating of the Marth and Fox player respectively, and  $\beta$  represents the effect rating differences has on the matchup. Running this new model using the same data from Section 3.2 produces the following results:

|                           | mean   | sd    | hdi_2.5% | hdi_97.5% | mcse_mean | mcse_sd | ess_bulk | ess_tail | r_hat |
|---------------------------|--------|-------|----------|-----------|-----------|---------|----------|----------|-------|
| alpha                     | -0.284 | 0.020 | -0.322   | -0.244    | 0.000     | 0.000   | 6584.000 | 2632.000 | 1.000 |
| beta                      | 0.002  | 0.010 | -0.018   | 0.022     | 0.000     | 0.000   | 6185.000 | 2853.000 | 1.000 |
| gamma[YOSHIS_STORY]       | 0.120  | 0.042 | 0.041    | 0.203     | 0.001     | 0.001   | 5576.000 | 2730.000 | 1.010 |
| gamma[FOUNTAIN_OF_DREAMS] | -0.018 | 0.047 | -0.107   | 0.073     | 0.001     | 0.001   | 7786.000 | 3162.000 | 1.000 |
| gamma[FINAL_DESTINATION]  | 0.068  | 0.047 | -0.023   | 0.156     | 0.001     | 0.001   | 7329.000 | 2771.000 | 1.000 |
| gamma[POKEMON_STADIUM]    | -0.081 | 0.044 | -0.172   | 0.001     | 0.001     | 0.001   | 6830.000 | 2919.000 | 1.000 |
| gamma[BATTLEFIELD]        | -0.019 | 0.043 | -0.106   | 0.065     | 0.001     | 0.001   | 7269.000 | 3220.000 | 1.000 |
| gamma[DREAMLAND]          | -0.070 | 0.047 | -0.156   | 0.023     | 0.001     | 0.001   | 7406.000 | 3410.000 | 1.000 |

Looking at the data for  $\beta$ , it appears that a difference in Slippi rating has little impact on the projected matchup. This is almost certainly due to the elo data being randomly generated; in fact, it likely serves as an example to show that the model is recognizing rating as a “random” factor of sorts. Given real, live data, I would expect this parameter to have the potential of changing.

For the  $\alpha$  and  $\gamma_i$  parameters, little has changed, also likely due to  $\beta$  having little impact on the matchup. While the matchup factor would likely remain the same, I would expect the stage effects to potentially diverge as skill difference is accounted for.

Although there is no substantial conclusion on player ratings impacting the Marth-Fox matchup in this example, this model provides the foundation to perform similar analysis in the future if match data containing player ratings is available.



# **Chapter 4**

## **Conclusion**

## 4.1 Conclusion

Using the Binomial-Beta model with the Slippi data and varying priors, we determined that the Marth-Fox matchup is close enough to be considered even. This contrasts Hjelte's claim that the Marth-Fox matchup is heavily favored for Marth. With the various logistic regression models, using a subset of the Slippi data, we determined that Yoshi's Story is definitively a strong stage for Marth against Fox. While not definitive, Final Destination appears to be in Marth's favor, while Dream Land and Pokemon Stadium appear to lean in Fox's favor. Battlefield and Fountain of Dreams appear to be relatively even. Final Destination is often regarded as Marth's best stage in the matchup, so it was interesting to see the average Marth player perform better on Yoshi's Story, relatively speaking. Finally, we adapted the logistic regression model to factor in the relative difference of player Slippi ratings when determining matchup probability. Although this data was not available in the dataset, future Slippi match data containing player ratings could use such data in this model for analysis.

There are an infinite number variables impacting the results of fighting games. A matchup may be awful on paper, but a player may practice it to such a degree that they make seem as though their character wins a matchup. Additionally, in competitive Melee, no two games are truly independent. Players use information about their opponents playstyles, habits, and skills to make strategic, on-the-fly adaptations. The free-flowing nature of a game like Melee results in infinite possibilities, and it is a main contributor to why its competitive scene has not only survived, but thrived over the last 25 years.

For the scope of this project, I opted to make assumptions about match independence and select a few major matchup factors to simplify the creation of a logistic regression model. There are several steps to improve upon the existing model. First, the model could utilize actual tournament data, where we could identify win probabilities given the result of the current match count in a best-of-five set. Additionally, data could be polled for specific Marth players to determine their success rate against specific Fox players or Fox players in general. A similar approach could be used for other characters in the cast. We could also run the model using data from only top players to determine if matchups change at different skill levels, or we could introduce an interaction effect to see if a combination of skill difference and stage selection work together to affect a matchup. Analysis of data like this could improve the viewer experience by showing live probabilities of a player winning given the current state of a set, or could be used to predict winners of major tournaments. Furthermore, this model could be extended to all 351 possible matchups in Melee, allowing for a more statistically-sound tier list, based on a combination of expert beliefs and observed data. Like Melee itself, there are limitless ways to leverage Bayesian methods to uncover something new about a beloved game.

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